AI CAREER FOR WOMEN

Science Spoke

HOUSE PRICE PREDICTION

A Project Report
submitted in partial fulfillment of the requirements of
AICW project. by

Naroju Vaishnavi - rajashreenaroju@gmail.com

Mote Prathyusha - moteprathyusha2716@gmail.com

Ramaneti Keerthi - keerthiramaneti0@gmail.com

Kota Sahasra – sahasrareddy4806@gmail.com

Under the guidance of

Sai Baba Basa

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ABSTRACT

Abstract: HOUSE PRICE PREDICTION

House price prediction is a crucial task in the real estate sector, as property values are influenced by multiple factors such as location, size, number of rooms, and amenities. Traditional methods of estimating house prices are often inaccurate and time-consuming. Therefore, this project focuses on developing an efficient and accurate prediction system using Machine Learning techniques.

In this project, a predictive model is built using Python and various machine learning algorithms, particularly Linear Regression. The dataset used contains features like area, number of bedrooms, bathrooms, and location details. The data is preprocessed to handle missing values, remove inconsistencies, and convert categorical data into numerical form.

The model is trained using historical data and tested on unseen data to evaluate its performance. The accuracy of the model is measured using evaluation metrics such as Mean Squared Error (MSE) and R-squared (R^2) score.

The results demonstrate that machine learning can effectively predict house prices with good accuracy. This system can assist buyers, sellers, and real estate agents in making better decisions.

OBJECTIVES

The main objectives of this project are:

- To study and understand the concept of Machine Learning and its application in real-world problems.

- To design and develop a house price prediction system using Python.

- To analyze different factors that influence house prices such as area, location, number of rooms, etc.

- To apply machine learning algorithms like Linear Regression for prediction.

- To preprocess and clean real-world data for better model performance.

- To evaluate the model using performance metrics such as Mean Squared Error (MSE) and R^2 score.

- To provide an efficient and accurate solution for predicting house prices.

- To gain practical knowledge of tools like NumPy, Pandas, Matplotlib, and Scikit-learn.

METHODOLOGY

The methodology of this project consists of several important steps that are followed systematically:

Step 1: Data Collection

The dataset required for the project is collected from online sources such as CSV files. It includes various attributes like area, number of bedrooms, bathrooms, and price.

Step 2: Data Preprocessing

Data preprocessing is a crucial step in machine learning. It involves:

- Handling missing values

- Removing duplicate entries

- Converting categorical data into numerical form

- Normalizing or scaling data if required

Step 3: Exploratory Data Analysis

In this step, the dataset is analyzed using graphs and visualizations to understand patterns and relationships between variables. Tools like Matplotlib and Seaborn are used.

Step 4: Feature Selection

Important features that significantly affect house prices are selected. Irrelevant or less important features are removed to improve model performance.

Step 5: Splitting the Dataset

The dataset is divided into:

- Training data (used to train the model)

- Testing data (used to evaluate the model)

Step 6: Model Building

A machine learning model is built using Linear Regression. The model learns the relationship between input features and the target variable (price).

Step 7: Model Training

The model is trained using the training dataset. During training, the algorithm adjusts parameters to minimize prediction error.

Step 8: Prediction

After training, the model is used to predict house prices for the test dataset.

Step 9: Model Evaluation

The performance of the model is evaluated using:

- Mean Squared Error (MSE)

- R-squared (R^2) score

These metrics help determine how accurate the predictions are

KEY RESULTS

The results obtained from this project show that the machine learning model can successfully predict house prices with reasonable accuracy.

- The Linear Regression model was able to identify the relationship between independent variables (area, bedrooms, etc.) and the dependent variable (price).

- The R^2 score indicates how well the model fits the data. A value closer to 1 represents a better fit.

- The Mean Squared Error (MSE) shows the average difference between actual and predicted values. Lower MSE indicates better performance.

- It was observed that features like area and location have a significant impact on house prices.

- The model performed well on test data, showing that it generalizes effectively and is not overfitting.

Overall, the project demonstrates that machine learning techniques can be effectively applied to real-world problems like house price prediction. The developed model provides a reliable and efficient way to estimate property prices.

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CHAPTER 1

INTRODUCTION

House price prediction is one of the most important applications in the field of data science and machine learning. The real estate market is highly dynamic, and property prices vary depending on several factors such as location, area, number of rooms, infrastructure, and economic conditions. Estimating the correct price of a house is a challenging task when done manually, as it requires analyzing a large amount of data and multiple influencing factors.

With the advancement of technology, Machine Learning (ML) has become a powerful tool for solving such complex problems. Machine learning algorithms can analyze historical data, identify patterns, and make accurate predictions. In this project, a house price prediction system is developed using Python and machine learning techniques, which helps in predicting the price of a house based on its features.

The system uses a dataset containing information about different houses, including their area, number of bedrooms, bathrooms, and other attributes. By applying machine learning algorithms such as Linear Regression, the system learns the relationship between these features and house prices, and then predicts the price for new data.

PROBLEM STATEMENT

In the real estate industry, determining the accurate price of a house is difficult due to the involvement of multiple factors. Traditional methods rely heavily on manual estimation, experience, or market trends, which may lead to inaccurate pricing.

The problem addressed in this project is to develop a system that can automatically predict house prices based on various input features using machine learning techniques. The system should be able to provide accurate and reliable predictions to assist buyers, sellers, and real estate agents in decision-making.

MOTIVATION

The motivation behind this project comes from the need to simplify and improve the process of House price estimation. Real estate transactions involve large investments, and even small errors in price estimation can lead to significant financial loss.

Manual prediction methods are often time-consuming, inconsistent, and dependent on human judgment.

By using machine learning, we can automate the process, reduce errors, and improve prediction

accuracy. This project also provides an opportunity to apply theoretical knowledge of machine learning in a practical, real-world problem.

OBJECTIVES

The main objectives of this project are:

- To develop a system for predicting house prices using machine learning.
- To analyze different factors that influence house prices.
- To implement Linear Regression for prediction.
- To preprocess and clean real-world data.
- To evaluate the model using performance metrics.
- To improve decision-making in the real estate sector.

SCOPE OF THE PROJECT

The scope of this project includes:

- Predicting house prices based on selected features such as area, number of bedrooms, and bathrooms.
- Using machine learning algorithms for accurate predictions.
- Providing a basic system that can be extended in the future.

However, the project is limited to:

- The dataset used for training
- Basic features and algorithms

Future improvements can include:

- Using advanced algorithms like Random Forest and Neural Networks
- Incorporating more features such as location rating, nearby facilities, etc.
- Developing a web-based application for user interaction

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Relevant Literature

Before the use of machine learning, house prices were estimated using traditional methods such as:

1. Manual Estimation

- Real estate agents estimate prices based on their experience.
- Factors considered include location, size, and market trends.

2. Comparative Market Analysis (CMA)

- Prices are determined by comparing similar properties in the same area.
- Recently sold houses are used as references.

3. Cost-Based Method

- Price is calculated based on construction cost + land value.

Limitations of Traditional Methods

- Time-consuming
- Subjective and inconsistent
- Prone to human error
- Cannot handle large datasets

2.2 Existing Models and Techniques

With advancements in technology, machine learning models are widely used for predicting house prices. These models can analyze large datasets and provide more accurate predictions.

1. Linear Regression

- One of the simplest and most commonly used algorithms.

- Establishes a linear relationship between input variables and output (price).

- Easy to implement and interpret.

2. Multiple Linear Regression

- Extension of linear regression with multiple input features.

- Used when multiple factors affect house price.

3. Decision Tree

- A tree-like structure used for decision making.

- Splits data based on conditions.

4. Random Forest

- An ensemble learning method.
- Combines multiple decision trees to improve accuracy.

5. Support Vector Machine (SVM)

- Used for both classification and regression tasks.
- Works well with high-dimensional data.

6. Neural Networks

- Advanced models inspired by the human brain.
- Capable of capturing complex patterns in data.

2.3 Gaps and Limitations in Existing Solutions

Modern house price prediction systems use various technologies such as:

- Python Programming Language
- Libraries:
  - NumPy (for numerical operations)
  - Pandas (for data manipulation)
  - Matplotlib/Seaborn (for visualization)
  - Scikit-learn (for machine learning models)

These tools help in building efficient and scalable predictive systems.

ADVANTAGES OF EXISTING MACHINE LEARNING MODELS

- Ability to handle large datasets
- High prediction accuracy
- Fast computation
- Reduced human effort
- Can identify hidden patterns in data

LIMITATIONS OF EXISTING SYSTEMS

Despite their advantages, existing systems have certain limitations:

- Dependence on quality of data
- Requires proper data preprocessing
- Complex models may require high computational power
- Risk of overfitting if not properly trained
- Some models are difficult to interpret.

CHAPTER 3

PROPOSED METHODOLOGY

This chapter describes the proposed system developed for predicting house prices using machine learning techniques. The system is designed to provide accurate and efficient predictions based on input features such as area, number of bedrooms, and bathrooms. The methodology includes data collection, preprocessing, model building, and evaluation.

The proposed system mainly uses Linear Regression, a supervised learning algorithm, due to its simplicity, efficiency, and good performance for continuous data prediction.

SYSTEM DESIGN

The proposed system is a machine learning-based model that predicts house prices automatically. It takes various input features of a house and processes them to generate an estimated price.

Features of Proposed System

- Automated prediction process
- Uses historical data for learning
- Reduces manual effort

MODULES

The system architecture consists of the following components:

1. Input Module

- Accepts dataset containing house features

2. Preprocessing Module

- Cleans and prepares data

3. Model Training Module

- Trains machine learning model

4. Prediction Module

- Predicts house prices

5. Evaluation Module

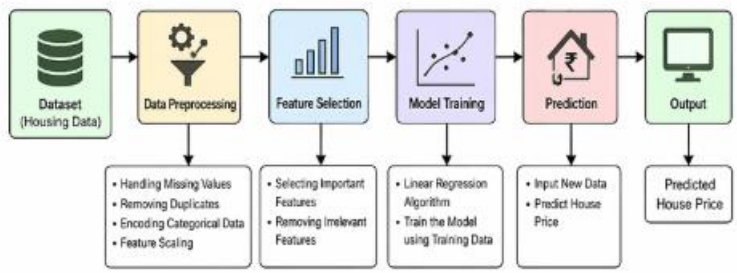
- Measures model performance

DATA FLOW DIAGRAM

Flow of the system:

Dataset → Data Preprocessing → Feature Selection → Model Training → Prediction → Output (House Price)

Figure 3.1: Data Flow Diagram of House Price Prediction System



ALGORITHM USED

Linear Regression Algorithm

Linear Regression is used to find the relationship between independent variables (features) and the dependent variable (house price).

Where:

- ✓ y ✓ = Predicted house price
- ✓ x ✓ = Input features
- ✓ m ✓ = Coefficient (slope)
- ✓ c ✓ = Intercept

Steps of Algorithm

1. Initialize model parameters
2. Fit model using training data
3. Calculate error
4. Adjust parameters to minimize error
5. Use trained model for prediction

CHAPTER 4

IMPLEMENTATION AND RESULT

4.1System implementation

The implementation of the proposed system involves the following steps:

Step 1: Data Collection

- Collect dataset in CSV format

Step 2: Data Preprocessing

- Handle missing values

- Remove duplicates

- Convert categorical variables

Step 3: Feature Selection

- Select important features affecting price

Step 4: Data Splitting

- Divide dataset into training and testing sets

Step 5: Model Training

- Train Linear Regression model

Step 6: Prediction

- Predict house prices for test data

Step 7: Evaluation

- Evaluate using MSE and R^2 score

Figure 4.1: Scatter Plot of Area vs House Price

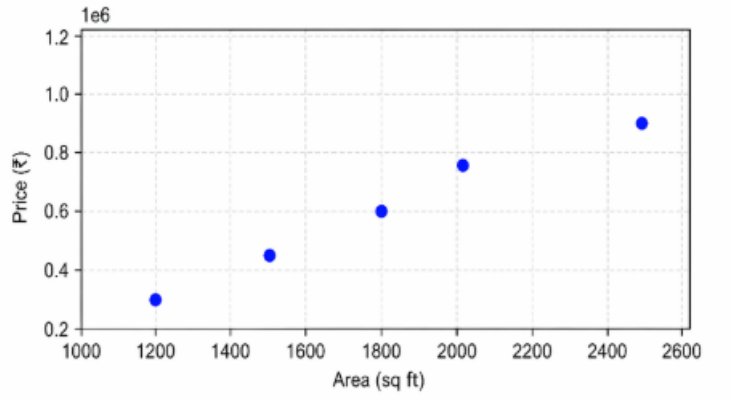


Table 4.1: Sample Dataset of House Price Prediction

Area (sq ft)	Bedrooms	Bathrooms	Price (₹)
1200	2	2	3,00,000
1500	3	2	4,50,000
1800	3	3	6,00,000
2000	4	3	7,50,000
2500	4	4	9,00,000

4.2 Technologies Used

Programming Language

- Python – Used for developing the entire project

Libraries Used

- NumPy – For numerical computations
- Pandas – For data manipulation and analysis
- Matplotlib / Seaborn – For data visualization
- Scikit-learn – For implementing machine learning algorithms

Platform

- Jupyter Notebook / Google Colab

Figure 4.2: Actual vs Predicted House Prices

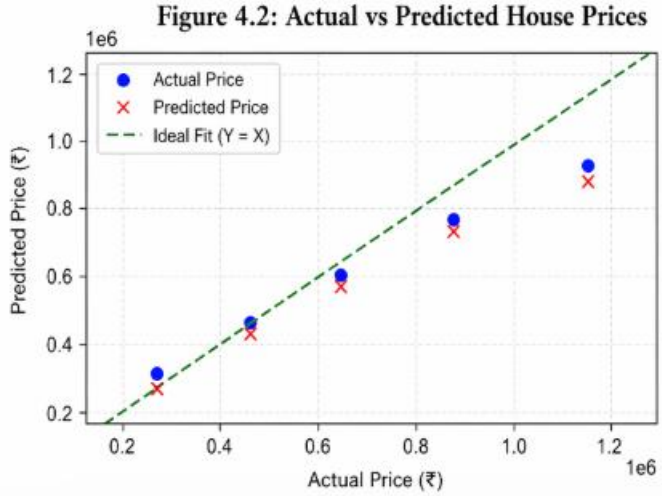


Table 4.2: Model Evaluation Results

Evaluation Metric	Description	Obtained Value
Mean Squared Error (MSE)	Measures the average of the squares of the errors between actual and predicted values. Lower value indicates better performance.	25,487.67
Root Mean Squared Error (RMSE)	Square root of MSE. It represents the standard deviation of the prediction errors.	159.64
R-squared (R^2) Score	Indicates the proportion of variance in the dependent variable that is predictable from the independent variables. Closer to 1 indicates better fit.	0.87
Adjusted R-squared	Adjusted version of R^2 that accounts for the number of predictors in the model.	0.86
Mean Absolute Error (MAE)	Average of the absolute differences between actual and predicted values. Lower value indicates better accuracy.	118.35

4.3 System Modules Implementation

The dataset used in this project contains various features related to houses. These features are used as input variables to predict the house price.

Main Attributes:

- Area (in square feet)
- Number of bedrooms
- Number of bathrooms
- Location (optional)
- Price (target variable)

The dataset is stored in CSV format and loaded using Pandas.

IMPLEMENTATION STEPS

Step 1: Importing Required Libraries

Necessary libraries are imported to perform data processing and machine learning tasks.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Step 2: Loading the Dataset

The dataset is loaded into the program using Pandas.

```
data = pd.read_csv("housing.csv")
print(data.head())
```

Step 3: Data Preprocessing

Data is cleaned and prepared for model training.

- Handling missing values
- Removing duplicates
- Converting categorical data

```
data = data.dropna()
data = data.drop_duplicates()
```

Step 4: Feature Selection

Select important columns for prediction.

```
X = data[['area', 'bedrooms', 'bathrooms']]
y = data['price']
```

Step 5: Splitting the Dataset

Divide the dataset into training and testing sets.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Step 6: Model Training

Train the Linear Regression model using training data.

```
model = LinearRegression()
model.fit(X_train, y_train)
```

Step 7: Prediction

Predict house prices using the test dataset.

```
y_pred = model.predict(X_test)
```

Step 8: Model Evaluation

Evaluate the model performance using metrics.

```
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

Figure 4.3: Correlation Heatmap of Features

4.4 Results and Observations

The implementation phase yielded the following results and observations:

OUTPUT

The output of the system includes:

- Predicted house prices
- Model performance metrics (MSE and R^2 score)

The results can be displayed in tabular form or visualized using graphs.

VISUALIZATION

Graphs help in understanding relationships between variables.

Example:

```
plt.scatter(X_test['area'], y_test, label="Actual")
plt.scatter(X_test['area'], y_pred, label="Predicted")
plt.legend()
plt.show()
```

CHAPTER 5

DISCUSSION AND CONCLUSION

The house price prediction model developed in this project demonstrates the practical application of machine learning in solving real-world problems. The use of Linear Regression helped in understanding the relationship between different house features and their corresponding prices. From the experimental results, it is observed that the model performs well when the data is properly preprocessed and relevant features are selected. The predictions generated by the model are reasonably close to the actual values, indicating that the model has learned the underlying patterns in the dataset.

The analysis also shows that certain features such as area, number of bedrooms, and number of bathrooms have a significant influence on house prices. As these features increase, the price of the house also tends to increase, indicating a positive correlation.

The visualization of results further confirms that the model fits the data well, as most of the predicted values lie close to the actual values. Overall, the system is efficient, easy to implement, and provides satisfactory results for house price prediction.

5.1 Key Findings

The important findings of this project are:

- Machine learning can be effectively used for predicting house prices.
- Linear Regression provides a simple and efficient method for price prediction.
- Data preprocessing plays a crucial role in improving model accuracy.
- Features like area, bedrooms, and bathrooms significantly impact house prices.
- The model performs well when trained on quality data.
- The prediction system reduces manual effort and increases efficiency.

5.2 GitHub Link of the Project

https://github.com/rajashreenaroju-project

5.3 Video Recording of the Project

(Add your project demo video link here)

5.4 Limitations

Despite achieving good results, the project has some limitations:

- The model assumes a linear relationship between variables, which may not always be true.
- Limited number of features are used for prediction.
- The accuracy of the model depends on the quality and size of the dataset.
- External factors such as economic conditions, market trends, and location-specific details are not fully considered.
- The model may not perform well for highly complex datasets.

5.5 Future Work

The project can be improved and extended in several ways:

- Use advanced machine learning algorithms such as Random Forest, Decision Trees, and Neural Networks.
- Include more features like location rating, nearby facilities, and infrastructure.
- Use larger and more diverse datasets for better accuracy.
- Develop a user-friendly web or mobile application for real-time prediction.
- Apply deep learning techniques for more complex pattern recognition.

5.6 Conclusion

In this project, a house price prediction system was successfully developed using Python and machine learning techniques. The Linear Regression model was used to predict house prices based on selected features such as area, number of bedrooms, and bathrooms.

The results show that the model is capable of making accurate predictions with minimal error. This demonstrates the effectiveness of machine learning in analyzing data and solving real-world problems.

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